23AID498 – Project Phase 1

Project Diary

Student Name:	Amitha . K . Anilkumar
Roll No:	AM.AI.U4AID23028
Team No:	15
Guide Name:	Shalu Murali
Project Title:	Data chain AI : Blockchain based secure marketplace for AI dataset sharing

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Project Diary (To be maintained by all students)

Date of meeting with the Guide: 23/06/26

Work planned for the coming week	• Perform an in depth review of the selected research papers • Identify research gaps and define the novelty of proposed system. • Refine the problem statement and finalize project objectives
Work completed during the previous week	• Finalised the project topic and identified problem domain • Conducted an initial literature survey by reviewing relevant research papers • understood topic objectives and scope.
Pending work	• Complete the initial project proposal document • Obtain guide's review and incorporate the suggested modifications.

Remarks by the Guide:

nil

Weekly Performance Rating (0–5): 5 / 5

Guide's Signature with Date:

Malu

23/06/26

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 10/07/26

Work planned for the coming week	• Study baseline models and perform a comparative analysis of existing approaches • Understand the outputs and limitations of current systems • Finalize the proposed system architecture and module design
Work completed during the previous week	• Reviewed additional research papers related to blockchain based AI dataset marketplace • Studied existing methodologies baseline models, and identified potential research gaps • Analyzed different approaches for dataset quality evaluation
Pending work	• Complete the review of the remaining research papers • Finalize the literature survey and research gap analysis • Begin designing the implementation workflow.

Remarks by the Guide:

nil (they were busy with CIR training)

Weekly Performance Rating (0–5): 4 / 5

Guide's Signature with Date:

Shaly

13/7/26

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Project Diary (To be maintained by all students)

Date of meeting with the Guide: 18/07/26

Work planned for the coming week	• Identify and finalize the AI models and algorithms required for each module. • Design the overall project pipeline and system blueprint. • Study suitable Explainable AI techniques and determine the most appropriate approach.
Work completed during the previous week	• Conducted an extensive review of journal papers. • Identified a key research gap : existing systems focus on secure ownership, provenance and transactions but lack explainable AI - based mechanisms to evaluate dataset trustworthiness
Pending work	• Review additional research papers to gain deeper understanding on blockchain.

Remarks by the Guide: nil

Weekly Performance Rating (0–5): 5 / 5

Shalu 18/7/26.

Guide's Signature with Date:

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 25/07/2026

Work planned for the coming week	• Study and implement an existing research paper recommended by the project guide to understand its implementation, workflow and technologies, which will serve as the foundation for developing the proposed framework
Work completed during the previous week	• Finalized the project objectives research gap, novelty and expected contributions. • Prepared the literature review presentation • Researched Agentic AI and identified its potential role.
Pending work	• Finalize the complete system architecture and workflow of the proposed Datachain AI framework • Identify and collect suitable public datasets for implementation

Remarks by the Guide:

→ Need to finalize system architecture.

→

Weekly Performance Rating (0–5): 5 / 5

Guide's Signature with Date: Shalu 25/7/26.

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Project Diary (To be maintained by all students)

Date of meeting with the Guide: 5/8/2026.

Work planned for the coming week	• Prepare for mid evaluation presentation • continue debugging the DQSM implementation after the evaluation. • Resume implementation once the library issues are resolved
Work completed during the previous week	• Studied and analyzed the implementation methodology from the selected research paper. • Attempted to implement the proposed model based on the paper • Set up the required development environment.
Pending work	• The implementation could not be completed because DQSM library was not functioning correctly and code execution resulted in error.

Remarks by the Guide: Shalu Murali need to work more.
Shalu Murali
5/8/2026

Weekly Performance Rating (0–5): 4 / 5

Guide's Signature with Date:

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Project Diary (To be maintained by all students)

Date of meeting with the Guide: 19/08/26

Work planned for the coming week	→ Improve the execution efficiency of the multi-agent quality assessment system → Implement parallel execution of 7 dataset quality assessment agents through the orchastrator.
Work completed during the previous week	→ Reviewed and correction the existing dataset quality assessment agents → worked on improving the individual agent functionalities. → Attempted to implement parallel execution
Pending work	→ complete and validate parallel execution 7 agents → Evaluate execution time and ensure that the agents produce consistent assessment results.

Remarks by the Guide: Do the implementation

Weekly Performance Rating (0–5): 5 / 5

Guide's Signature with Date: Shalu

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Project Diary (To be maintained by all students)

Date of meeting with the Guide: 14/09/26

Work planned for the coming week	→ Adding Escrow to the model → completing the Blockchain part.
Work completed during the previous week	→ Implemented IPFS code → Did VSRP model code → Done evaluation using agentic AI
Pending work	→ Implementing VSRP → connecting IPFS to blockchain

Remarks by the Guide: Shalu Shalu Marrale 14/9/2026 Try to do implementation

Weekly Performance Rating (0–5): 5 / 5

Guide's Signature with Date:

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Project Diary (To be maintained by all students)

Date of meeting with the Guide: 22/09/26

Work planned for the coming week	-> Explain and improve the explainability component of dataset quality assessment system. -> Develop mechanisms to provide a better understanding of contents
Work completed during the previous week	-> worked on implementing explainability techniques for the dataset quality assessment results.
Pending work	-> Improve the accuracy & clarity of the generated dataset descriptions. -> Further integrate explainability techniques with the quality assessment and reposting modules.

Remarks by the Guide: nil

Weekly Performance Rating (0–5): 5 / 5

Guide's Signature with Date: Thalu

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Project Diary (To be maintained by all students)

Date of meeting with the Guide: 15/10/26

Work planned for the coming week	→VSRP implementation. Implement the module.
Work completed during the previous week	→Implemented IPFS functionality
Pending work	→ Gain better understanding on VSRP.

Remarks by the Guide: code Review completed

Weekly Performance Rating (0–5): 5 / 5

Guide's Signature with Date: Shalu 15/10/26.

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Project Diary (To be maintained by all students)

Date of meeting with the Guide: 28/10/26

Work planned for the coming week	→ Improve the IPFS based dataset storage and retrieval mechanism. → Enhance the marketplace escrow functionality to ensure secure transactions between buyers and sellers.
Work completed during the previous week	→ Reviewed the existing IPFS integration → Worked on improving escrow mechanism.
Pending work	→ Further improve the reliability of IPFS operations. → Complete testing of the escrow workflow, including failed verification & disputed transactions

Remarks by the Guide: nil

Weekly Performance Rating (0–5): 5 / 5

Guide's Signature with Date: Shalu

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Project Diary (To be maintained by all students)

Date of meeting with the Guide: 4/11/26

Work planned for the coming week	→ Implement and integrate VSRP → Develop commit reveal feedback mechanism
Work completed during the previous week	→ Implemented fingerprint similarity based soft peer grouping. → Worked on the reputation management mechanism to reward or penalize participants
Pending work	→ Test the complete feedback, peer validation and reputation update workflow

Remarks by the Guide: Do the implementation

Weekly Performance Rating (0–5): 5 / 5

Guide's Signature with Date: Shalu

23AID498 – Project Phase 1

Project Diary (To be maintained by all students)

Date of meeting with the Guide: 9/11/26.

Work planned for the coming week	→ Integrate the individual modules into a unified system → Test the overall workflow of the dataset marketplace and identify remaining issues
Work completed during the previous week	→ Worked on integrating agentic quality assessment, explainability, IPFS and blockchain modules.
Pending work	→ Develop remaining frontend components and perform end-to-end system testing

Remarks by the Guide:

nil

Weekly Performance Rating (0–5): 5 / 5

Guide's Signature with Date: Thalu